

# TARGET-90

**JKBOSE CLASS-11<sup>TH</sup>**

## MOTION IN PLANE

One Shot

**PHYSICS**

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## Topics to be Covered

**A** Motion in Plane ✓

**B** Projectile Motion ✓

~~~~~→ 5m

**C** Uniform Circular Motion ✓

**D** Centripetal Acceleration ✓

~~~~~→ 5m

## ✓ **Motion in a Plane**

✓  
Scalar and vector quantities; position and displacement vectors, general vectors and their notations; equality of vectors, multiplication of vectors by a real number; addition and subtraction of vectors, Unit vector; resolution of a vector in a plane, rectangular components, Scalar and Vector product of vectors.  
 ✓ Motion in a plane, cases of uniform velocity and uniform acceleration projectile motion, uniform circular motion. ✓



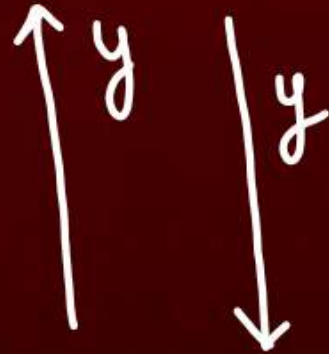


## Analyzing motion in plane

## Motion in 2D

motion in 1D | Rectilinear motion | Translational motion.

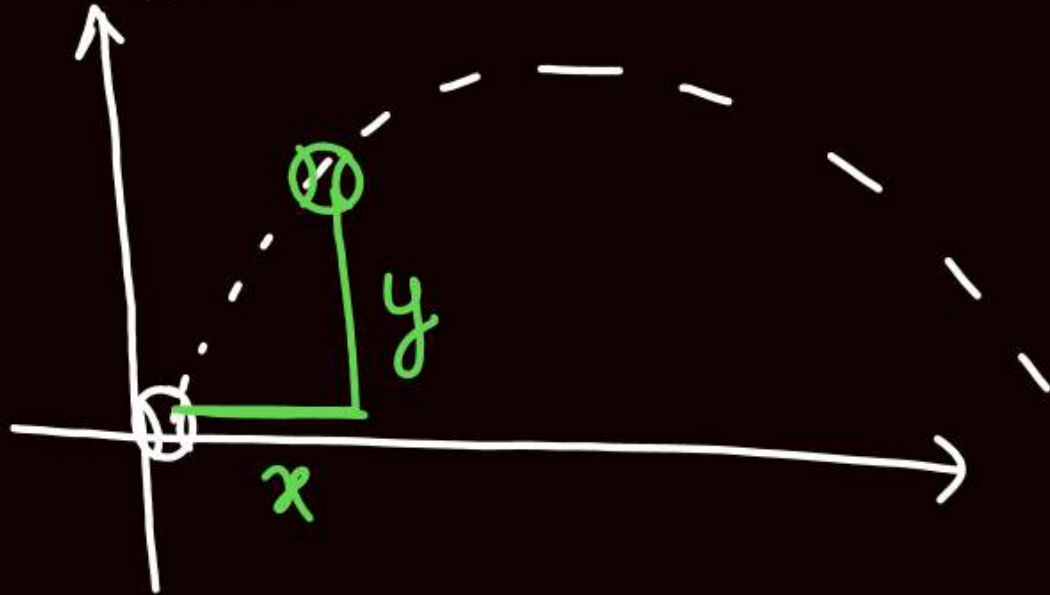
# When Body travel in  
Straight line



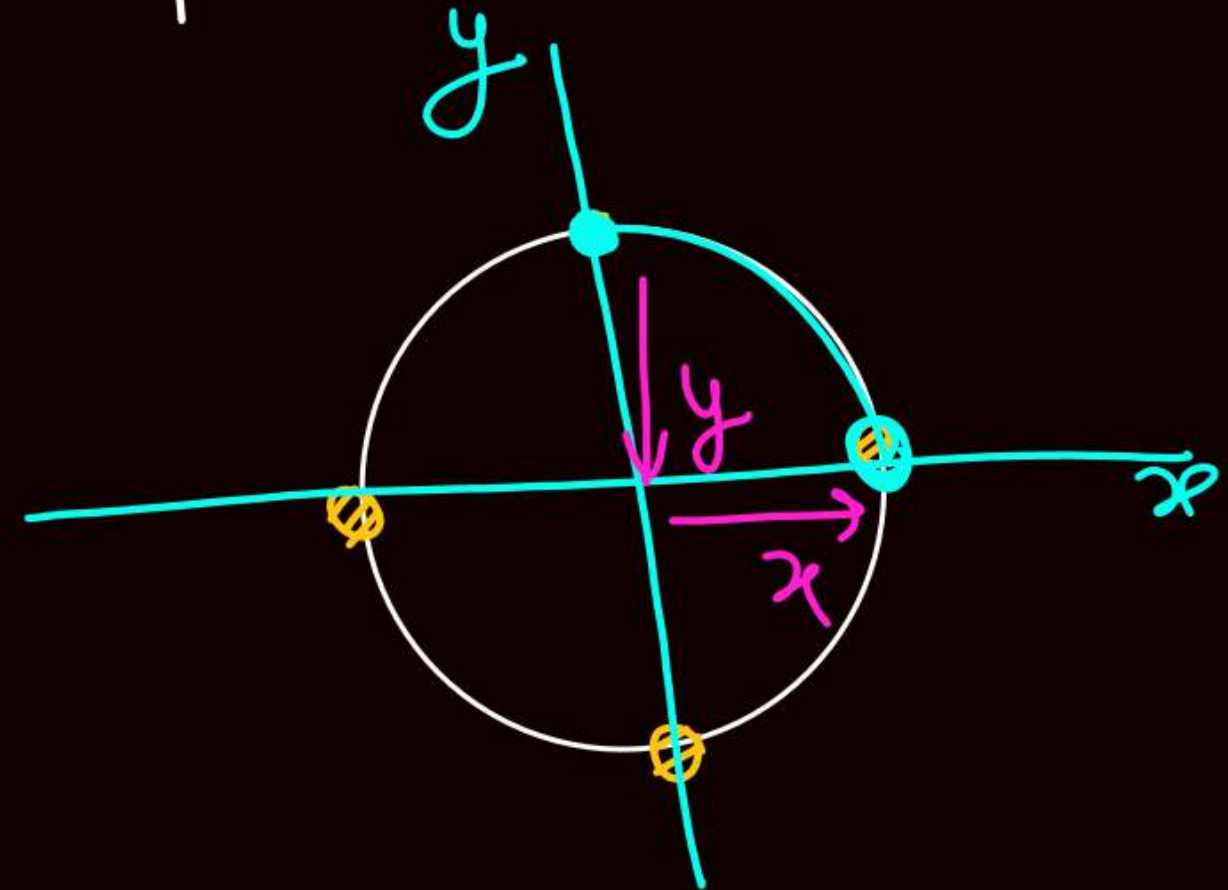
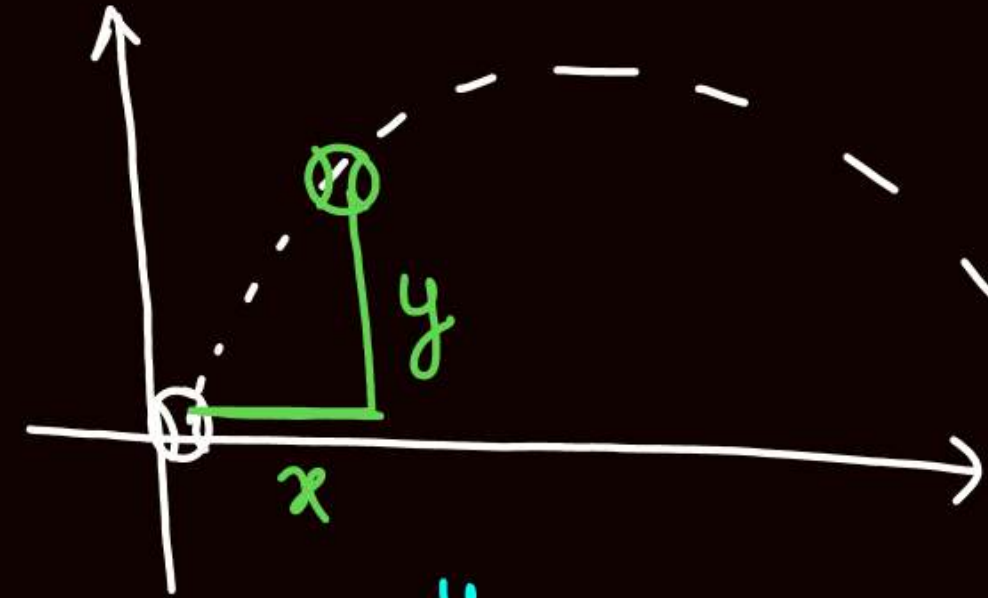
II

## Motion in 2D | Motion in plane

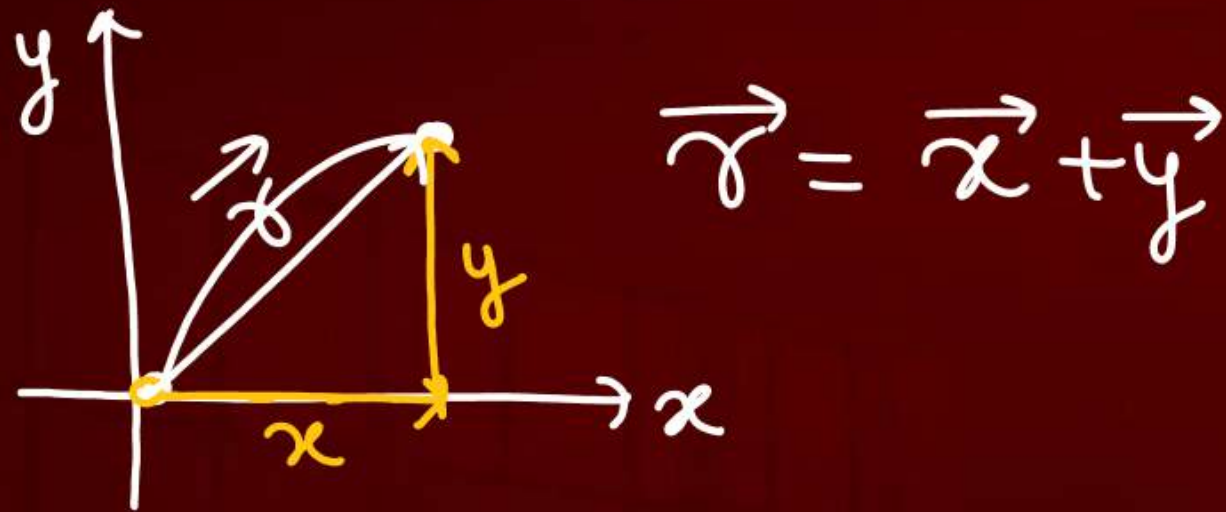
\* When Body moves in two direction at a Same



## (i) Projectile Motion



## Net Displacement 2D



## Net Velocity

The Rate of change of displacement  $\vec{r}$

$$\vec{r} = \vec{x} + \vec{y}$$

$$\frac{d\vec{r}}{dt} = \frac{d\vec{x}}{dt} + \frac{d\vec{y}}{dt}$$

$$\vec{v} = \vec{v}_x + \vec{v}_y$$

## Net Acceleration

# Rate of change of velocity

$$\vec{v} = \vec{v}_x + \vec{v}_y$$

$$\frac{d\vec{v}}{dt} = \frac{d\vec{v}_x}{dt} + \frac{d\vec{v}_y}{dt}$$

$$a = a_x + a_y$$



# Analyzing motion in plane

## Equations of Motion

$$V = u + at$$

$$V_x = u_x + a_x t$$

$$V_y = u_y + a_y t$$

$$S = Ut + \frac{1}{2} at^2$$

$$x = u_x t + \frac{1}{2} a_x t^2$$

$$y = u_y t + \frac{1}{2} a_y t^2$$

$$V^2 = u^2 + 2as$$

$$V_y^2 = u_y^2 + 2a_y y$$

$$V_x^2 = u_x^2 + 2a_x x$$



## Projectile Motion

2026  
2025

2024  
2023

2022  
2023

2021

# Example of  
2D-motion

Example of motion in plane

An object which is thrown in space with some initial velocity at an angle  $\theta$  with the horizontal under the influence of (g)

1. ✓ Time of flight

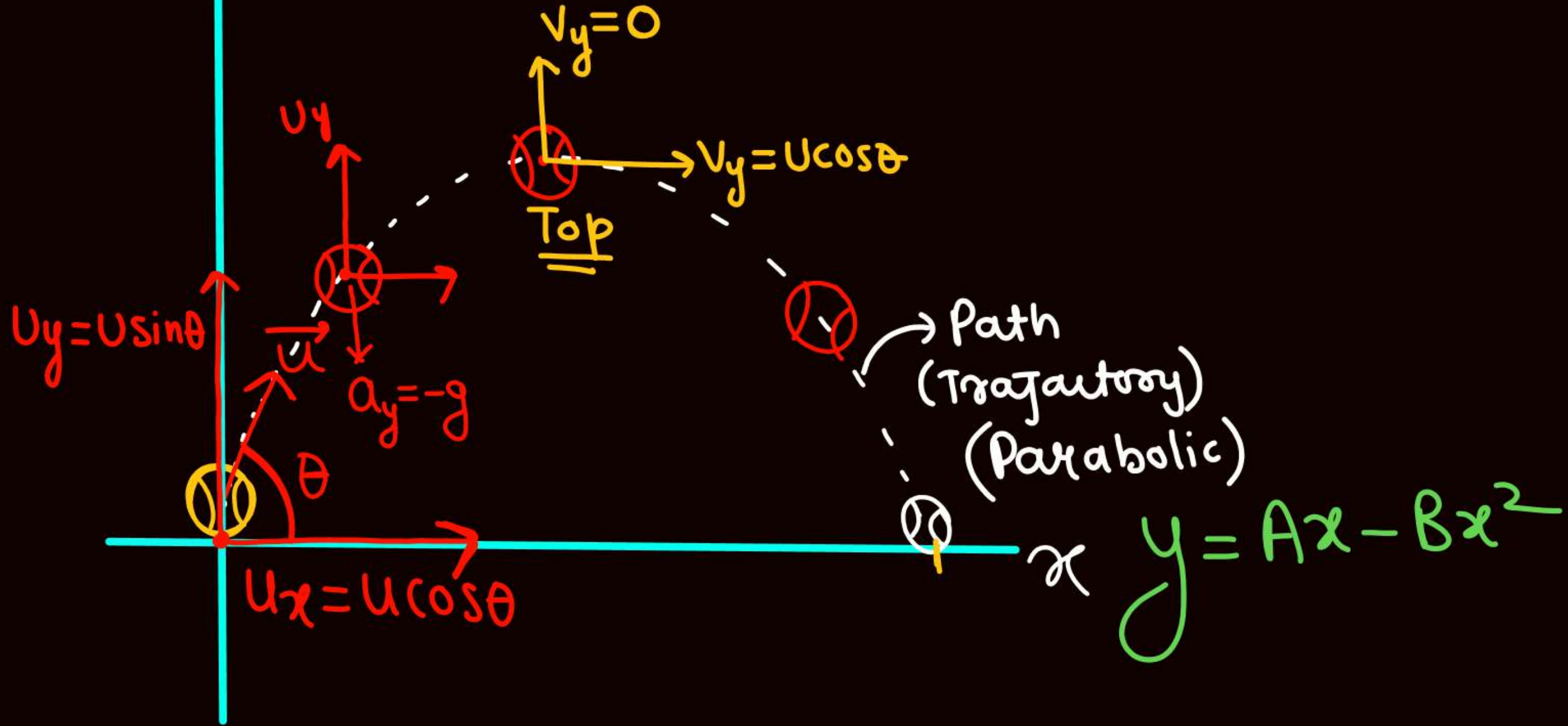
2. ✓ Maximum Height

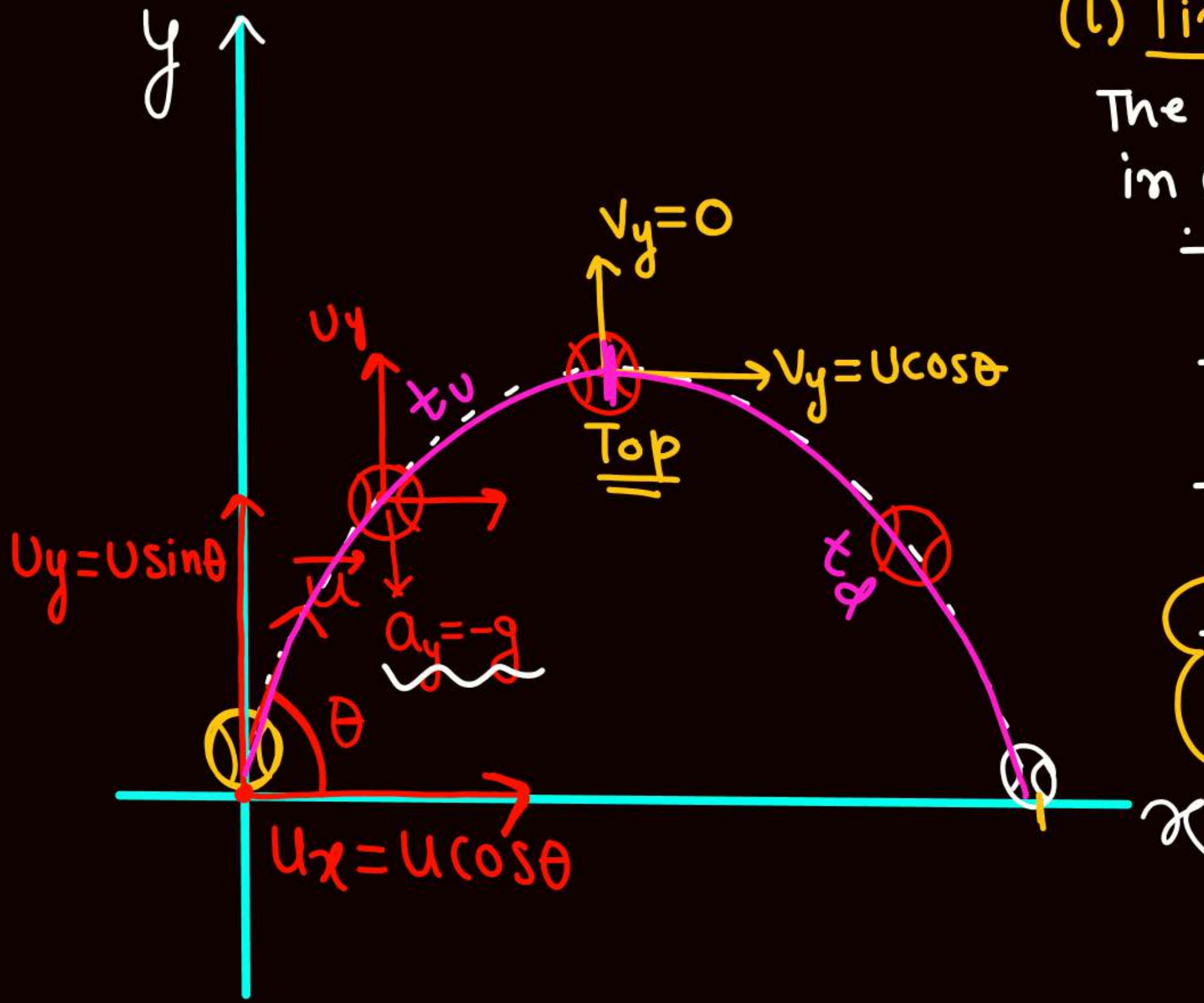
3. ✓ Range

4. Equation of Trajectory (x and y) (prove that the path of projectile is parabolic)

Imp

5 mark





## (i) Time of flight

The time for which ball remain in air

$$T = t_u + t_d$$

$$\underline{t_u} = \underline{t_d}$$

$$T = t_u + t_u$$

$$T = 2(t_u)$$

# From 1st equation

$$v_y = u_y + at$$

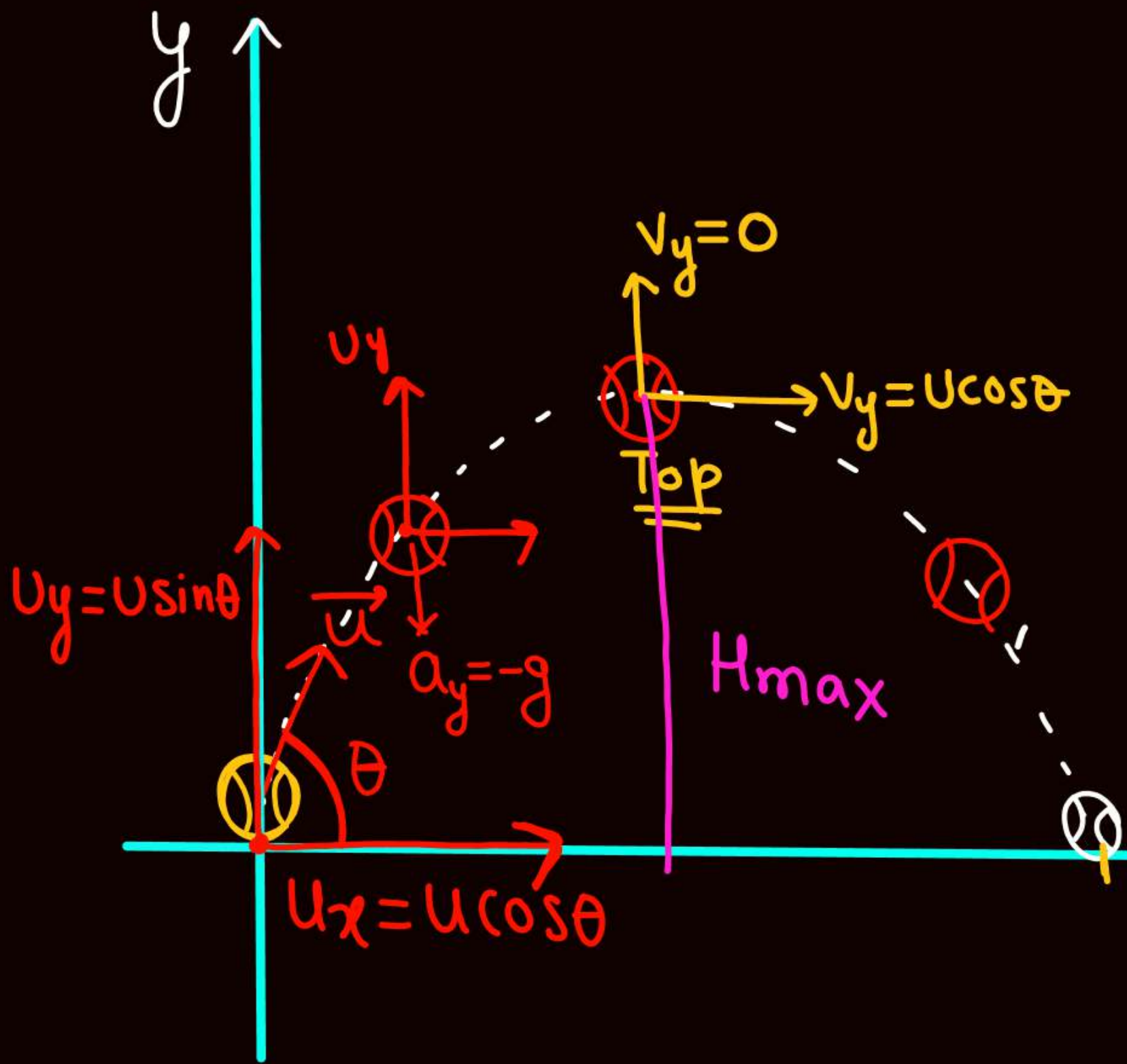
$$0 = u \sin \theta + (-g)t_u$$

$$0 = u \sin \theta - gt_u$$

$$gt_u = u \sin \theta$$

$$t_u = \frac{u \sin \theta}{g}$$

$$T = \frac{2u \sin \theta}{g}$$



## (2) Maximum Height

- The maximum vertical displacement is called Maximum Height

From 3rd equation

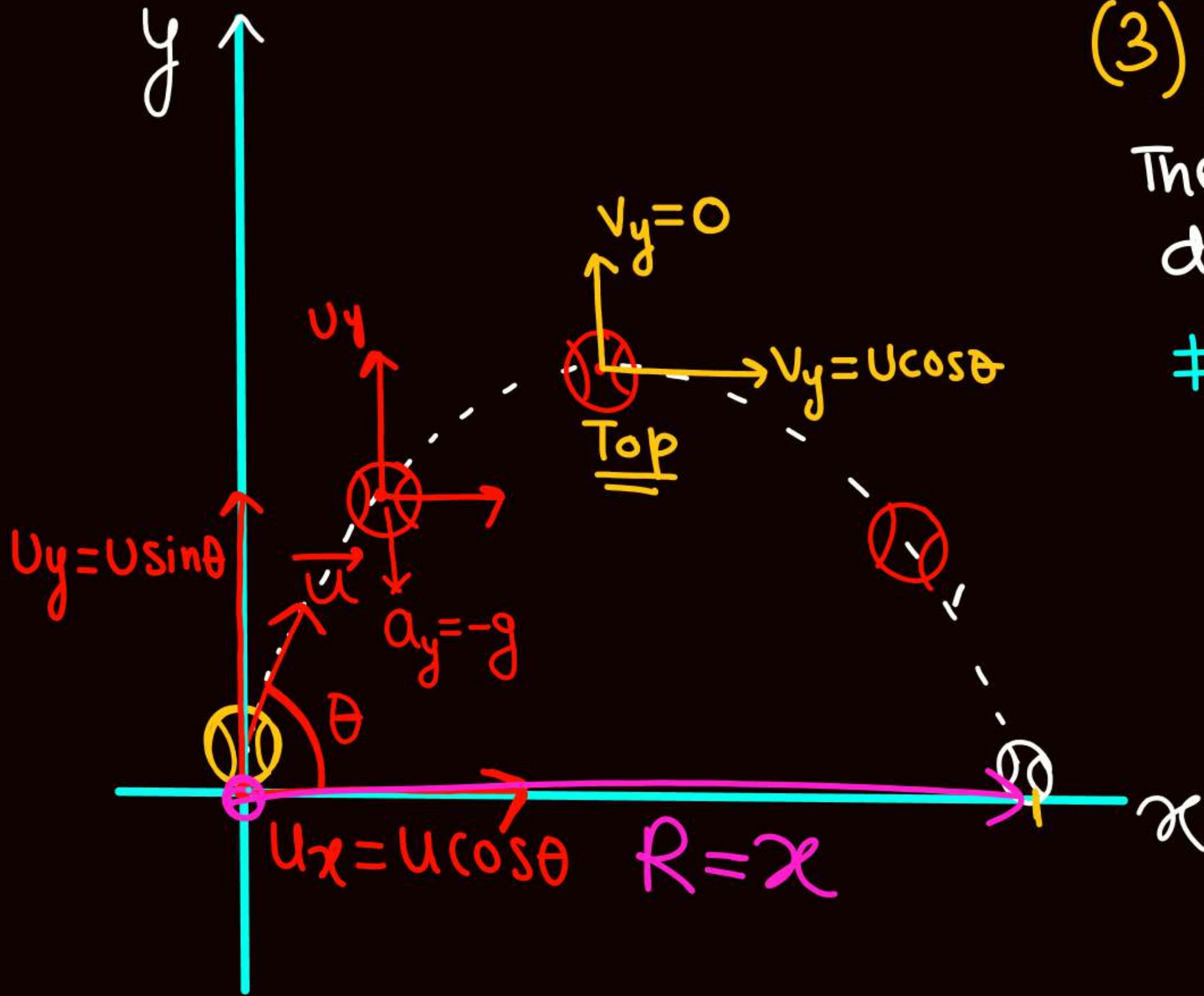
$$v_y^2 = u_y^2 + 2a_y y$$

$$0^2 = (u \sin \theta)^2 + 2(-g) H_{\max}$$

$$0 = u^2 \sin^2 \theta - 2g H_{\max}$$

$$\times 2g H_{\max} = u^2 \sin^2 \theta$$

$$H_{\max} = \frac{u^2 \sin^2 \theta}{2g}$$



### (3) Range

The maximum horizontal displacement

$$\# \quad V = \frac{x}{T}$$

$$u \cos \theta = R$$

$$\underline{T}$$

$$u \cos \theta \times T = R$$

$$R = u \cos \theta \underline{T}$$

$$R = u \cos \theta \frac{2u \sin \theta}{g}$$

$$T = \frac{2u \sin \theta}{g}$$



$$R = u \cos \theta + \frac{2u \sin \theta}{g}$$

$$R = \frac{u^2 (2 \sin \theta \cos \theta)}{g}$$

$$2 \sin \theta \cos \theta = \underline{\sin 2\theta}$$

$$R = \frac{u^2 \sin 2\theta}{g}$$

g

#### ④ Equation of trajectory

It is the Relationship b/w  $y$  and  $x$  - displacement  
# 2nd equation of motion

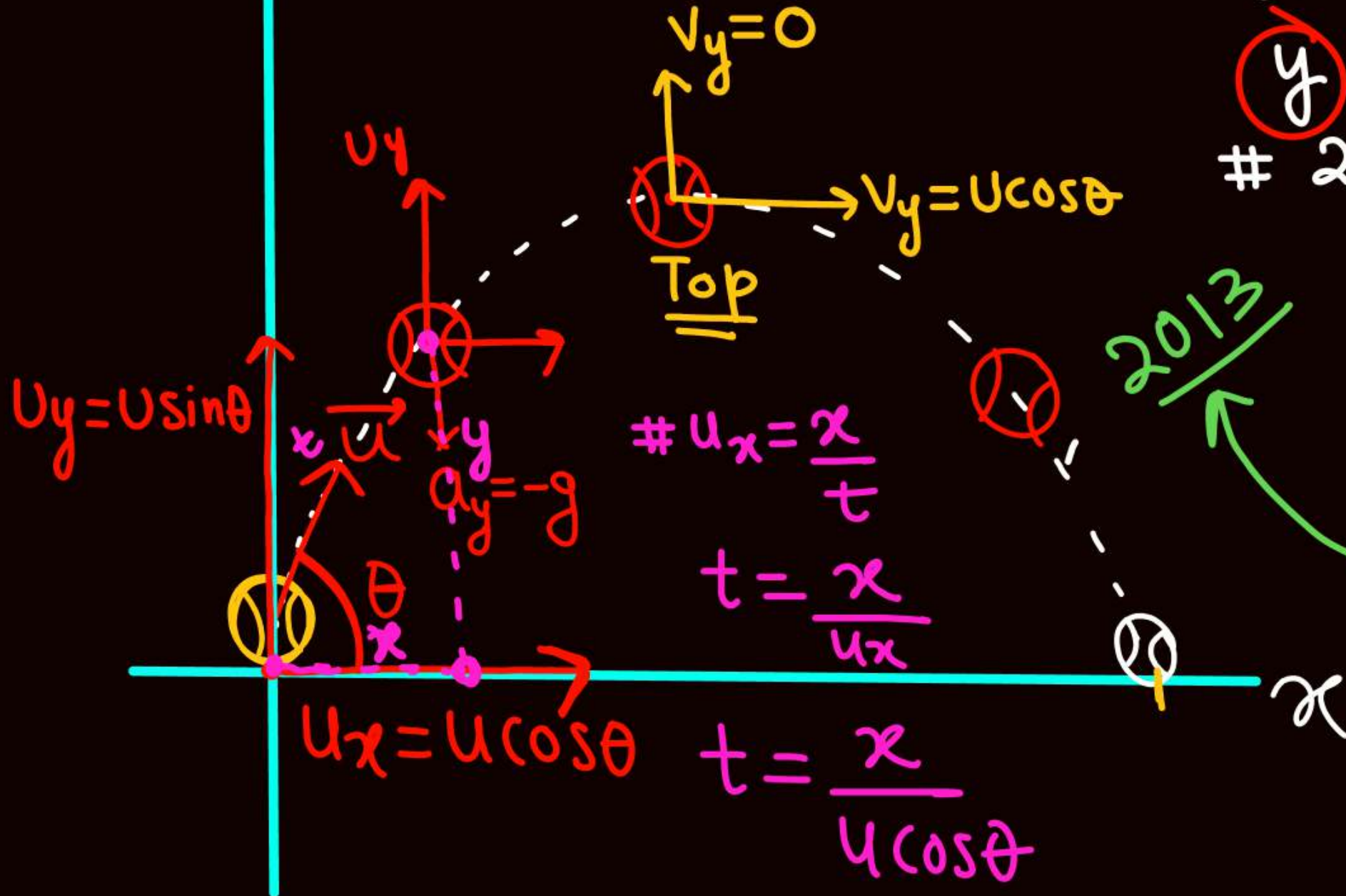
$$S_y = u_y t + \frac{1}{2} a_y t^2$$

$$y = u \sin \theta t + \frac{1}{2} (-g) t^2$$

$$y = u \sin \theta \left( \frac{x}{u \cos \theta} \right) - \frac{1}{2} g \left( \frac{x}{u \cos \theta} \right)^2$$

$$y = \cancel{u} \sin \theta \frac{x}{\cancel{u} \cos \theta} - \frac{1}{2} g \left( \frac{x}{\cancel{u} \cos \theta} \right)^2$$

$$y = x \frac{\sin \theta}{\cos \theta} - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$



$$y = x \frac{\sin \theta}{\cos \theta} - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$

$\tan \theta$

equation of trajectory

$$y = x \tan \theta - \frac{1}{2} g \frac{x^2}{u^2 \cos^2 \theta}$$

$$y = x \tan \theta - \frac{1}{2} \frac{g}{u^2 \cos^2 \theta} x^2 \rightarrow \text{Parabolic.}$$

$$y = x \underline{A} - B x^2$$

graph  $\rightarrow$  Parabolic





## → Speed constant Uniform Circular Motion

### # Example of 2D motion

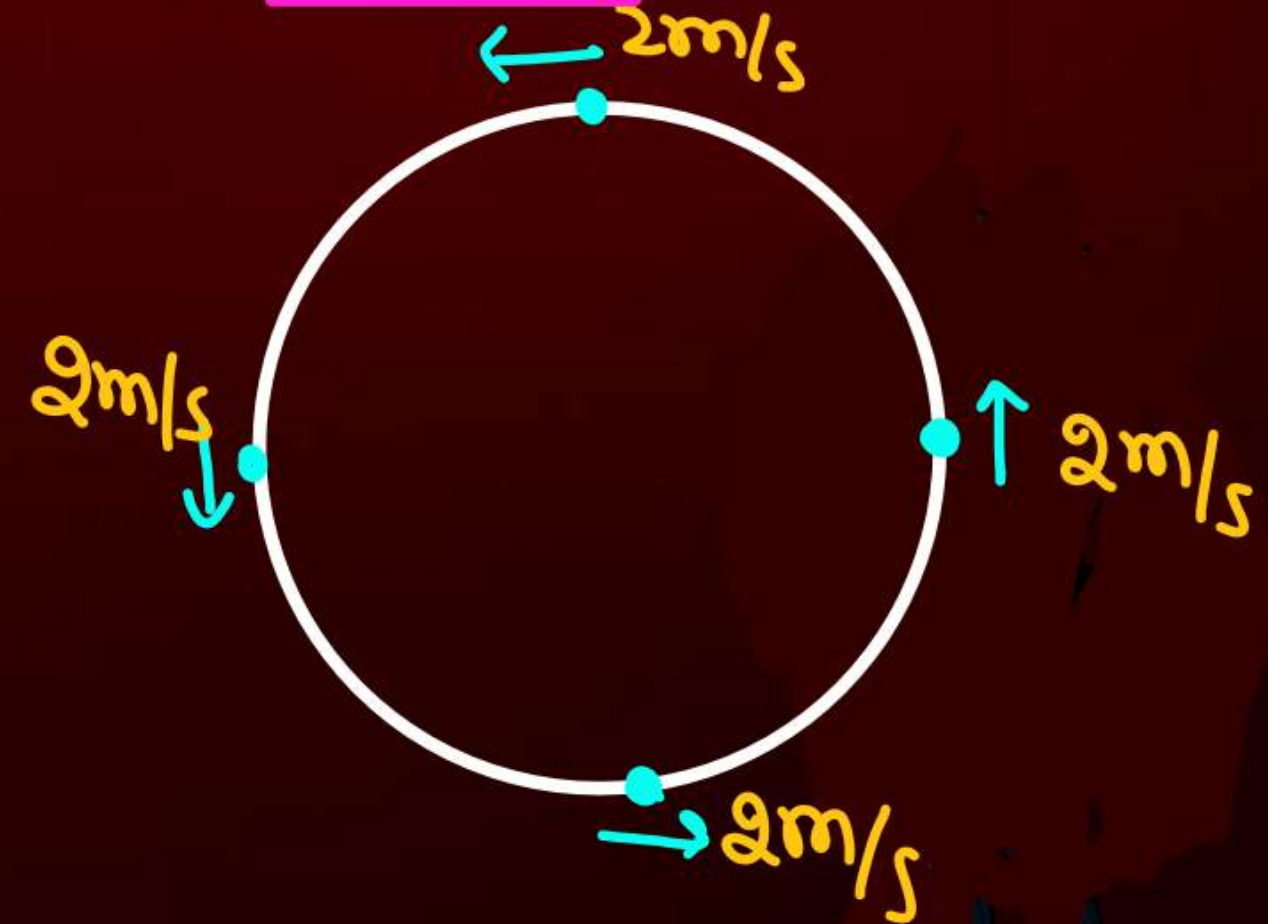
The movement of the body following a circular path is called circular motion.

Now the motion of a body moving with constant speed along a circular path is called **Uniform Circular Motion**. Here, the speed is constant but the velocity changes.

MCQ Speed =  $2\text{m/s}$  (magnitude) constant

Change Velocity =  $\underbrace{2\text{m/s}}_m \times \underbrace{\text{Direction}}_{\text{change}}$

non-uniform motion



Note :- The acceleration which change the Direction of velocity is called Centripetal acc

$$a_c = \frac{v^2}{R} \quad (\text{Derivation 5mark})$$



# Uniform Circular Motion

$$\theta = \underline{\underline{Rad}}$$

## 1. Angular Displacement ( $\theta$ )

# The Angle displaced by the Body During Circular motion

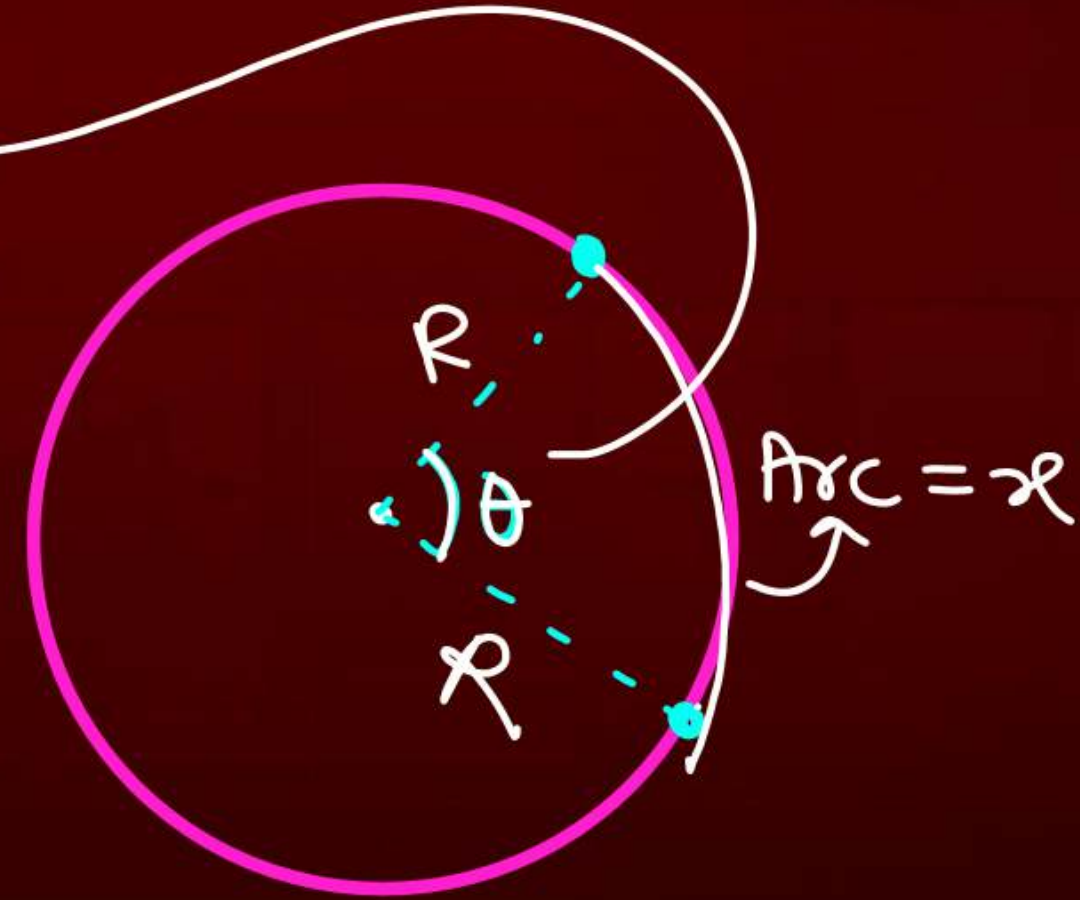
Angular displacement

$$\theta = \frac{\text{Arc}}{R}$$

$$\theta = \frac{x}{R}$$

$$R\theta = x$$

Linear  $x$





## Uniform Circular Motion

### 2. Angular Velocity ( $\omega$ )

# the Rate of change of Angular displacement is called Angular velocity

$$\omega = \frac{d\theta}{dt}$$

$$\omega = \frac{\text{rad}}{\text{sec}}$$

#  $x = R\theta$   
on differentiating b/s

$$\frac{dx}{dt} = \frac{d}{dt} R\theta$$

$$V = R \left( \frac{d\theta}{dt} \right) \omega$$

$$V = R\omega$$

Linear Velocity

Angular Velocity



## Uniform Circular Motion

### 3. Angular Acceleration ( $\alpha$ )

The rate of change of Angular velocity

$$\alpha = \frac{d\omega}{dt}$$

$$\alpha = \frac{\frac{\text{rad}}{\text{sec}}}{\text{sec}} = \frac{\text{rad}}{\text{sec}} \times \frac{1}{\text{sec}} = \frac{\text{rad}}{\text{sec}^2}$$

$$\# V = R\omega$$

on differentiating b/s

$$\frac{dV}{dt} = \frac{dR\omega}{dt}$$

$$a = R \left( \frac{d\omega}{dt} \right)$$

$$\swarrow \quad a = R\alpha \rightarrow \text{angular acc.}$$

linear acc

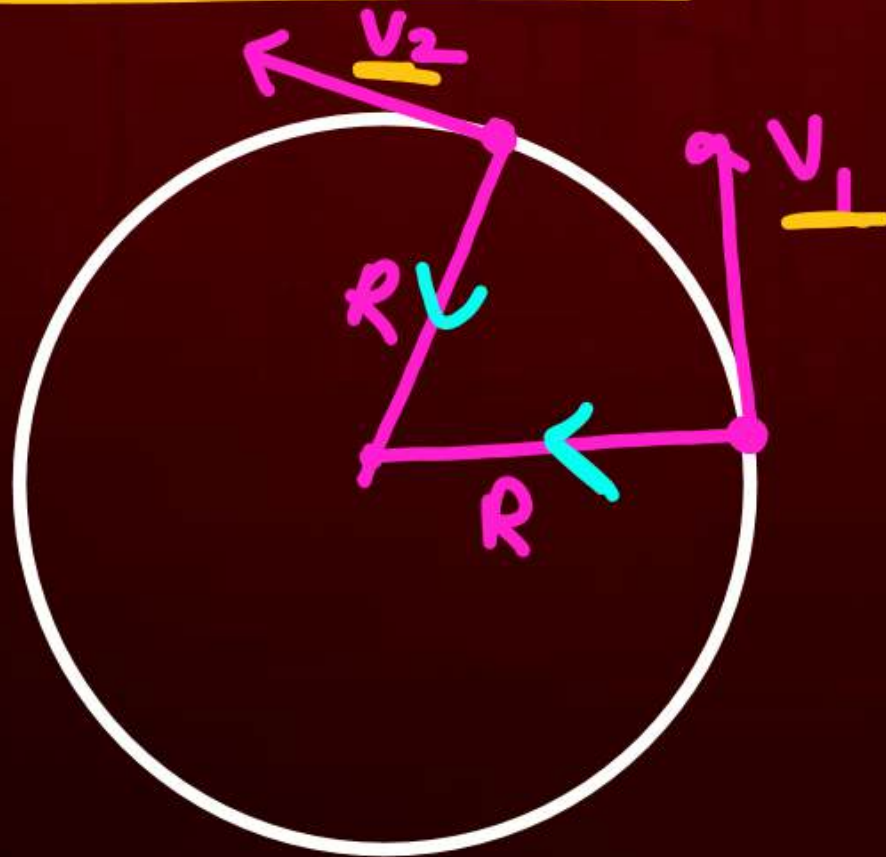


# Centripetal Acceleration

2025 2016  
2024 2018  
2022  
2014

The centripetal acceleration is the acceleration which only change the direction of velocity.

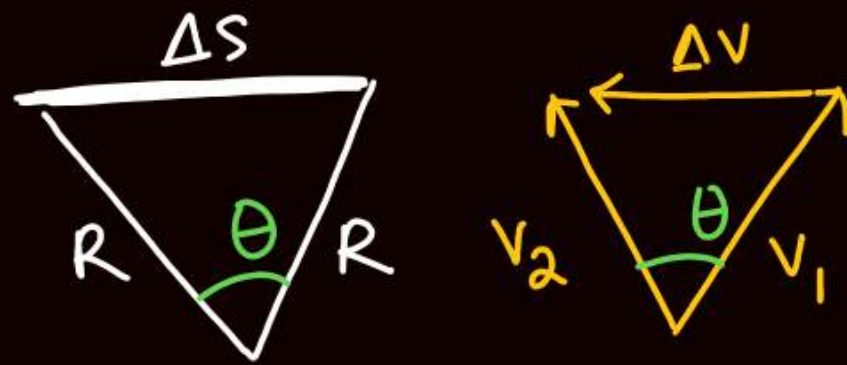
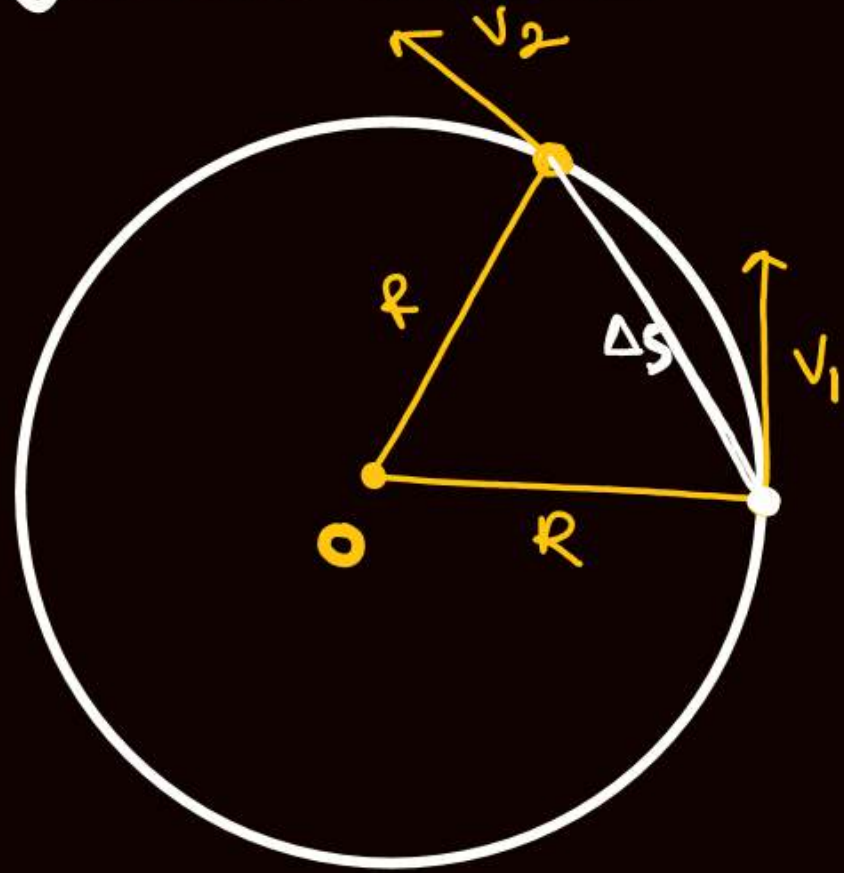
# Centripetal acc. always act towards the centre



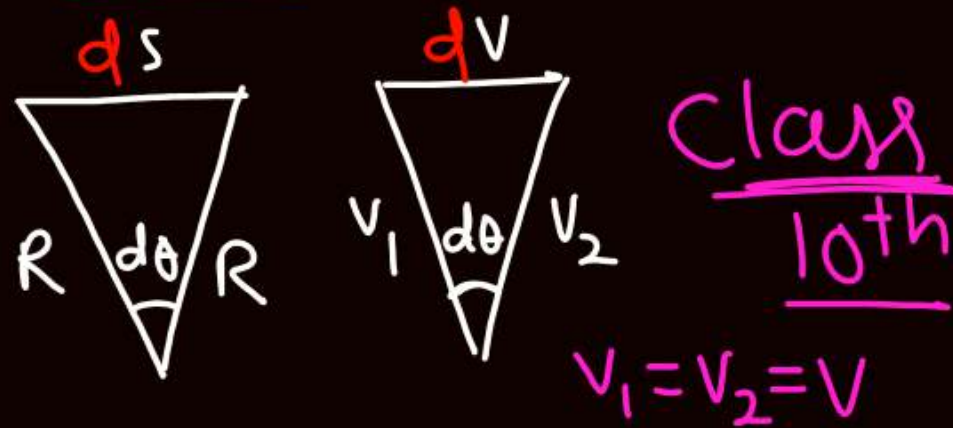
$$\# a_c = \frac{v^2}{R}$$



# Derivation



For Small Angle  $d\theta$   
Both the  $\Delta$  become  
Similar



$$\frac{ds}{R} = \frac{dv}{v}$$

$$\frac{ds}{R} = \frac{dv}{v}$$

Divide b/s by t

$$\frac{ds}{Rt} = \frac{dv}{vt}$$

$$\frac{v}{R} = \frac{a}{v}$$

$$\frac{v \times v}{R} = a_c$$

$$\frac{v^2}{R} = a_c$$



AAA

^

# Centripetal Force

# The force which  
change the  
direction of  
velocity

$$F_c = ma_c$$

$$F_c = m \frac{v^2}{R}$$





## Question

**The path traced by a projectile in space is known as:**

- A** Track
- B** Orbit
- C** Trajectory
- D** Coral



## Question

2025

5 marks

**Show that the path followed by projectile is parabolic when it is projected at an angle  $\theta$  with horizontal.**



## Question

2026  
2m



**Define projectile. Derive an expression for maximum height time of flight and horizontal range of a projectile fired at an angle with the horizontal.**



## Question

2024

**Find an expression for the maximum height time of flight and horizontal range of a projectile fired at an angle with the horizontal.**



## Homework



All derivation JKBOSE.

**Thank  
You**